# Production systems & planning

D&M WITH MR KELLY

Higher · Materials & manufacture · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ compare one-off and mass production
- ☐ explain why standard components and knock-down fittings are used
- ☐ describe batch production and the issues that decide the production system
- ☐ explain how automation, CAD/CAM, CNC, Gantt and flow charts, jigs and JIT improve production

## I DO — Worked example

Read how a full-mark answer is built

**1 Outline three benefits to a manufacturer of using standard components.** (3 marks)

**Full-mark answer** each ✓ is one mark

They are cheaper to buy in bulk than making the parts in-house ✓. They are made to a consistent quality, so the manufacturer knows they will work ✓. They are readily available, so fewer production stages are needed and production is faster ✓.

**Why answers lose marks**

*"Easy to replace if they break."*

0 marks: this is a benefit to the consumer, not the manufacturer.

## WE DO — Guided practice

Use the starters and word bank to help you

**2 Explain issues a manufacturer should consider before choosing a production system for a new kettle.** (3 marks)

**Sentence starters**

One reason is … because …
Another reason is … so …
Also, … which means …

**Word bank**

workforce · needed · machinery · investment · tooling · expected · demand · volume

Tip: 3 marks = 3 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** What is a drawback of just-in-time production?
☐ Large warehouses needed ☐ Cannot easily cope with a sudden increase in orders ☐ More waste ☐ Slower deliveries

**b)** Which system has the highest set-up cost but the lowest cost per item?
☐ One-off ☐ Batch ☐ Mass ☐ Prototype

**c)** How does a jig improve production?
☐ Removes need for accurate tools ☐ Reduces marking out and improves accuracy and repeatability ☐ Increases waste ☐ Replaces all workers

**d)** Which is a disadvantage of standard components for a manufacturer?
☐ Guaranteed quality ☐ Dependence on an outside supplier ☐ Easy to source ☐ Saves time

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Mass production** | ____ | **A** A single unique product made by highly skilled workers. |
| 2. **One-off production** | ____ | **B** A set quantity made at once; production can switch between products. |
| 3. **Batch production** | ____ | **C** Large volumes of identical products; automated; low skill; low unit cost. |
| 4. **Standard component** | ____ | **D** Device that guides a tool for accurate, repeatable work. |
| 5. **Just-in-time** | ____ | **E** Supplies arrive exactly when needed, so no large stock is held. |
| 6. **Jig** | ____ | **F** A part made in large numbers to an agreed standard (e.g. bolts, hinges). |

*A manufacturer of the chairs uses standard components.*

### 5 Explain the benefits and drawbacks to the manufacturer of using standard components. (4 marks)

**6 Describe how Gantt charts and just-in-time production are used in the development and manufacture of products.** (4 marks)

**7 Explain why production and planning systems are used in the development and manufacture of new products.** (8 marks)

### Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?